Carmen Aznar Mathonneau

Title: Measuring Surface Electromyography with Textile Electrodes in a Smart Leg Sleeve

Authors: F. Amitrano, A. Coccia , G. Pagano, A. Biancardi, G. Tombolini, V. Marsico, G. D'Addio (2024)

MASTER BIOMEDICAL DATA SCIENCE

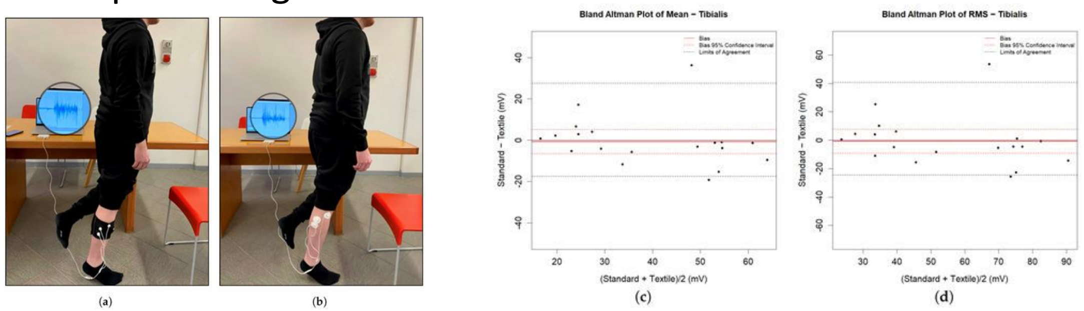
- Biomedical engineering: Wereable EMG sensors
- **Integration of sensor technology** (conductive textiles) in a **wearable biomedical device** and validation in human trial

Objective: To design and validate a **smart leg sleeve** equipped with **textile electrodes** (conductive fabric) for measuring surface electromyography (sEMG) from lower-limb muscles, and to compare its performance with conventional **Ag/AgCl gel electrodes**.

Methods: participants: 11 healthy volunteers; muscles: anterior tibialis and lateral gastrocnemius; parameters extracted: area, power, mean, RMS; statistical analysis: paired *t*-test + Bland–Altman plots; usability assessment: Comfort Rating Scale (CRS).

- Explain the contribution of the paper in respect other papers (state of the art).
- Perhaps it will be useful to fill in this section once make the global discussion of all the papers.

- No statistically significant differences between textile and Ag/AgCl electrodes for all metrics ($p>0.05$)
- Bland–Altman plots show $\mu \approx 0$ and a 95% limit of agreement ($\mu \pm 1.96$ SD), indicating strong correspondence between both systems.
- CRS reported high comfort levels.



- Study focused on static contractions, motion and sweating effects remain untested
- Small sample size and limited muscle selection
- Only healthy volunteers, no validation in patients with neuromuscular disorders or skin conditions
- Short-term testing:signal stability after repeated use or washing not assessed